

24 Multiple Reactions with Heat Effects

Why Are We Learning This?

In previous lectures, we developed energy balances for reactors involving a single reaction. In those systems, the heat generated by reaction was relatively straightforward to calculate because only one reaction contributed to the thermal behavior of the reactor. Real reactors are often more complicated. Desired and undesired reactions can occur simultaneously, intermediates may be formed and consumed, and different reactions may release or absorb heat at different rates. As a result, the reactor temperature is determined by the combined thermal effects of all reactions occurring in the system.

Today we extend the energy-balance concepts from single-reaction systems to reactors containing multiple reactions. By the end of this lecture, you should be able to:

- Write the heat-generation term for systems involving multiple reactions.
- Understand why each reaction contributes separately to the reactor energy balance.
- Construct non-isothermal reactor models involving multiple reactions.
- Recognize why conversion is generally insufficient for multiple-reaction energy-balance problems.
- Understand why numerical methods are usually required when temperature and multiple reactions are coupled.

Course Roadmap

Introduction to CRE



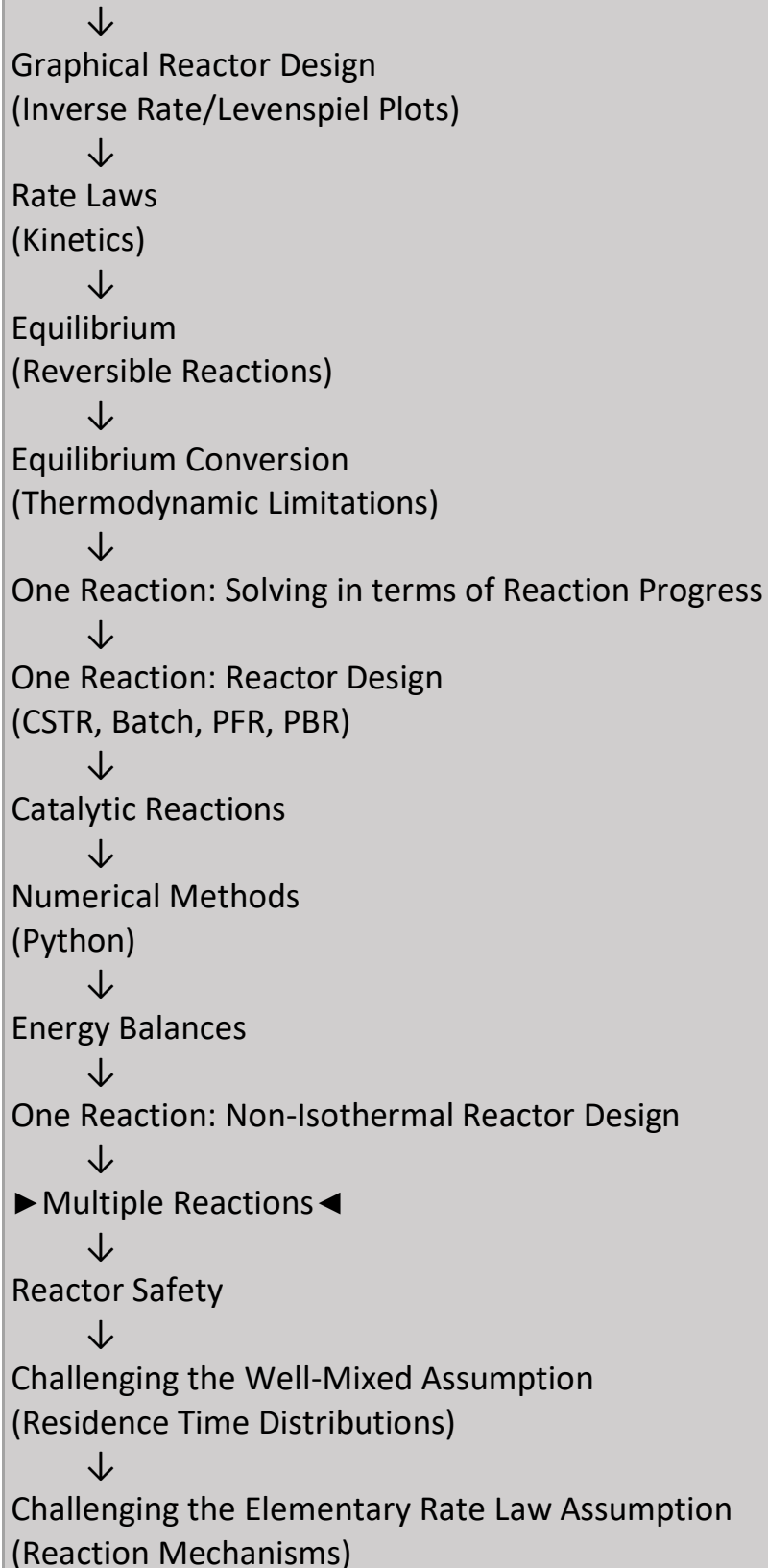
General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress
(Conversion)



Big Idea

Multiple Reactions



Multiple Heat Sources



Temperature Changes



Reaction Rates Change



More Temperature Changes

In non-isothermal systems, every reaction contributes to the reactor temperature, and the temperature influences every reaction.

Recall the energy balances that we've been looking at so far, where we have:

- Heat Generated term (Q_g or $G(T)$) -- due to the reaction
- Heat Removed term (Q_r or $R(T)$) -- due to the heat exchanger and also sensible heat changes as reactor contents "adsorb" heat given off by the reaction

The Heat Generated terms look like:

For the case of multiple reactions, the Heat Generated term looks like:

$\sum_i (-r_{ij})(-\Delta H_j^{rxn})$, where i are reactions and j are species.

In this sum, there should be **one $(-r_j)(-\Delta H_j^{rxn})$ term per reaction**. The purpose of the j subscript is to ensure consistent stoichiometry/units ... that each rate of reaction and heat of reaction pair are taken with respect to the same species. What matters here is that j is consistent for each pair; it is OK if j varies from pair to pair.

Why Doesn't One Heat-of-Reaction Term Work Anymore?

For a single reaction: Heat Generated = Reaction Rate $\times$ Heat of Reaction

Only one reaction contributes to heat generation.

For multiple reactions: Reaction 1 + Reaction 2 + Reaction 3 + ... all contribute simultaneously.

Therefore: Total Heat Generated = Sum of Heat Generated by Each Reaction

The reactor does not distinguish between desired and undesired reactions. Every reaction contributes to the thermal behavior of the system.

Exothermic and Endothermic Reactions Can Compete

Different reactions may have different heats of reaction. For example:

Reaction 1 = Exothermic

Reaction 2 = Endothermic

The total heat-generation term becomes: Heat Released - Heat Absorbed

As a result, the reactor temperature depends on the balance between all reactions occurring simultaneously.

Physical Interpretation

Reaction 1



Releases Heat

Reaction 2



Consumes Heat

Net Effect



Determines Reactor Temperature

This is one reason why multiple-reaction systems can behave very differently from single-reaction systems.

Beyond that, the solution procedure is analogous to the one you've already used to solve reactor design problems when there are multiple reactions: You have to use N 's or F 's instead of X 's, and because of this you need to use mathematical software.

Why Conversion Is Usually Not Enough

For a single reaction: One conversion variable can often describe the entire system.

For multiple reactions: Species may be produced, consumed, produced again, consumed again, etc. through different reactions. Because of this, a single conversion variable can no longer describe the complete state of the reactor.

Instead, we must track:

Moles (N) or Molar Flow Rates (F)

for individual species. This is why most multiple-reaction, non-isothermal reactor-design problems require numerical solution methods.

Let's do an example:

The gas phase elementary reactions $2A \rightleftharpoons B$ and $2B + A \rightarrow C$ are carried out in a non-isothermal PFR with no pressure drop. Write the symbolic expression for the heat generated term.

Info:

$$k_{1A} = 50 \text{ M}^{-1} \text{ min}^{-1} \text{ at } 350 \text{ K with } E = 8 \text{ kJ/mol}$$

$$k_{2C} = 4000 \text{ M}^{-2} \text{ min}^{-1} \text{ at } 310 \text{ K with } E = 4 \text{ kJ/mol}$$

$$K_{C,A} = 10 \text{ M}^{-1} \text{ at } 315 \text{ K}$$

$$\Delta H_{1A}^{rxn} = -20 \text{ kJ/mol A}$$

$$\Delta H_{1A}^{rxn} = +30 \text{ kJ/mol B}$$

Summary and Key Takeaways

Today we extended reactor energy balances from single-reaction systems to systems involving multiple reactions.

Key Ideas

- ✓ Every reaction contributes separately to the heat-generation term.
- ✓ The total heat generated is the sum of the heat generated or consumed by each reaction.
- ✓ Exothermic and endothermic reactions may compete within the same reactor.
- ✓ Multiple-reaction systems generally must be solved using species flow rates or moles rather than a single conversion variable.
- ✓ Temperature affects reaction rates, and reaction rates affect temperature.
- ✓ Numerical methods are often required because the mole balances and energy balance become strongly coupled.

Most Important Idea

Reaction 1



Heat Effect

Reaction 2



Heat Effect

Reaction 3



Heat Effect

Total Heat Generated



Reactor Temperature



Reaction Rates

In multiple-reaction non-isothermal reactors, the heat-generation term is no longer associated with a single reaction. The reactor temperature is determined by the combined thermal effects of all reactions occurring simultaneously.